STORMHACKS 2026 · TEAM HANDOUT

ATLAS Build Graph

Who builds what, what can run side by side, what has to wait, and the gates in between.

This is the build plan cut for people. Each box is one area of work that one person owns, about the size of “implement hyperbolic terrain generation”, not a single task. For every phase it shows the areas of each lane, which of them can start at the same time, which one waits for another, what is handed from one person to the next, and the gate each area feeds.

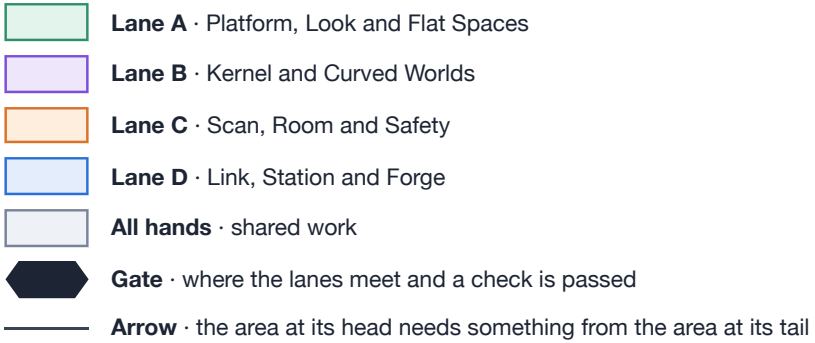
Everything on these pages is read from `documents/ATLAS_Master_Plan.md` when the PDF is built. The work areas are the ones the plan defines (§9.5.2, §9.6, §9.7, §9.8); who waits for whom is worked out from the dependencies of the tasks inside each area. The tasks themselves, with their outputs and acceptance tests, stay in the plan (§9 and §10) for the coding agents. Work is ordered by dependencies and gates only.

HOW TO READ AN AREA

- **Starts after** (on the boards: **after**): nothing inside the area can begin until the named area or gate is done.
- **Also needs** (**needs**): the area can start, but one part of it waits for something from another area.
- **Feeds**: the gates at which the area's work is checked.
- **Hands over to**: what another lane is waiting for from this area. These are the moments to talk.
- A tag such as `B2` is lane B, second area of that phase. Task ids such as `P1–B04` (phase 1, lane B, number 04) only point into the plan. Gates are G0 to G15.

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LEGEND



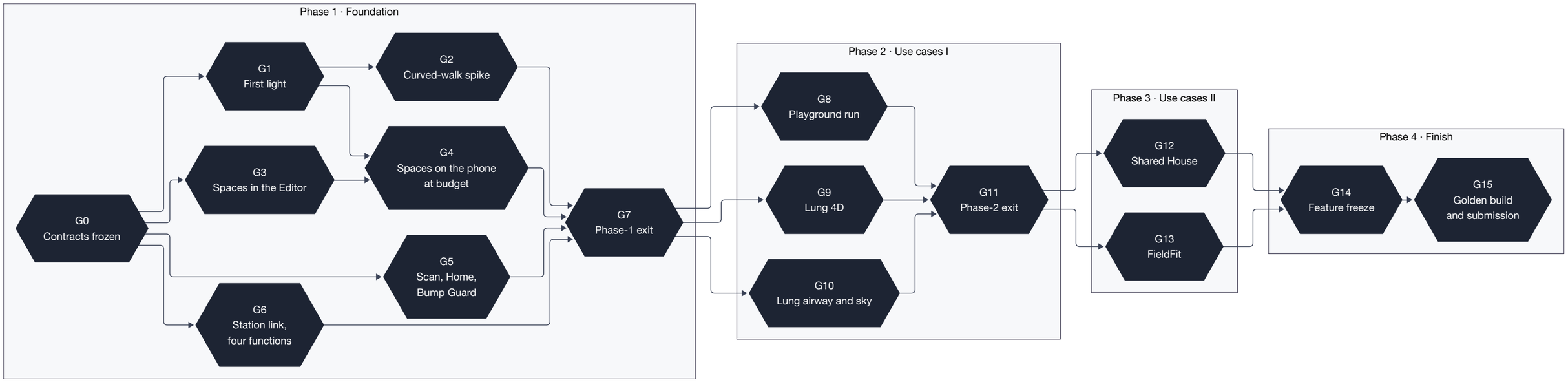
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1 The big picture

1.1 Phases and gates

plan §9.4



Gate	Passes when (summary; full checklists in §11.3)
G0 Contracts frozen	Every contract of §6.4 is written and stubbed; each lane's sandbox runs against the stubs
G1 First light	Empty stereo scene at 60 fps on the phone; telemetry visible on the Station; anyone can install a build
G2 Curved-walk spike	Five-corner loop closes within 0.25 m and three walkers from the team are comfortable at the loop-fit curvature (with a spotter: Bump Guard does not exist yet). Decides the default mode and the comfort floor
G3 Spaces in the Editor	S1–S6 run in SpaceLab on the three sample rooms; all mathematics tests green
G4 Spaces on the phone at budget	Every space S0–S6 meets the Must column cold, with the room's content, on the `budget_room` fixture; the 15-minute soak passes
G5 Scan, Home and Bump Guard	Both scan paths end in a registered Home; Bump Guard passes its scripted test in the headset
G6 Station link, four functions	Mirror, tuning and Console, telemetry, and the scan and asset round-trip all pass §7.6
G7 Phase-1 exit	A first-time walker goes from self-scan to every space, watched and tuned from the Station
G8 Playground run	A first-time walker completes the guided run in two different rooms
G9 Lung 4D	The four 4D pieces run on both case packages at budget
G10 Lung airway and sky	Four forks walked without rescue; sky and plan card correct
G11 Phase-2 exit	Playground and Lung Mission in one session by an outsider, hot, at budget
G12 Shared House	Post Circling walked by three people with no contact
G13 FieldFit	Blind spot found; search is faster with the lens than without
G14 Feature freeze	Full path twice in a row on one phone without cooling; only fixes after this
G15 Golden build and submission	Golden build runs with the Station unplugged; documents, video and metrics delivered

Phase 3's packages depend technically only on Phase 1. They are placed after G11 because every lane is occupied until then.

1.2 The four lanes: who owns what

plan §9.2

Lane	Phase 1 · Foundation	Phase 2 · Use cases I	Phase 3 · Use cases II	Phase 4 · Finish
A · Platform, Look and Flat Spaces	Project, build seat, stereo rig, lens blit, shared shader, SpaceLab, Glued (S1), Impossible Rooms (S5), Slicer and `Mesh4D` (S6), room content in Glued, thermal ladder, soak	Slicer toybox, equation sources, look of the Playground; the whole medical 4D runtime (worldtube, breath lane, scanner tilt, tasks) and the Lung director with its labels and credits	Shared House: sheets, door, second home, visitor	Look and audio, performance hardening, golden build
B · Kernel and Curved Worlds	Kernel, holonomy rig, warp shader, exact tile ids, tile meshes and renderer, world generator, Small Planet (S4), dial and 3-4-5 walk, Aligned Cells (S3), room as tile zero, room content in Aligned Cells	Lung Mission: airway wing in hyperbolic space, valve chambers, Two Compasses, needle sky	FieldFit: lens mathematics, lookup, composed final blit, blind-spot finder	Maths notes, correctness report, equation-card audit, figures
C · Scan, Room and Safety	AR session, FloorScan core (Editor), scan routes, RoomModel and loop-fit, Home (S0), object photos, Bump Guard with its warning cue and haptics, input, pose pipeline, onboarding, pre-scan path	Geometry Playground: the shell (director, guided walk, captions, twins, menu), the pointing probe, and the Glued and Impossible Rooms exhibits	Shared House: second scan, door fit, Bump Guard on the second sheet, peer link	Onboarding with outside walkers, comfort protocol, failure drills, rehearsals
D · Link, Station and Forge	`Atlas.Link`, Hub, tuner and Console, perf probe and telemetry, Spectator, flat-twin map, Forge and mesh clean-up, asset client, built-in fallback set, on-device menu, pairing	Lung Mission: the Station pipeline (`atlas case`), case packages, plan card, Lung panels on Console and Spectator	FieldFit: field presets, Console lens panel, Spectator heatmap, search task and logging; Shared House: peer relay (Extended) and two-plan flat twin	Judges' dashboard, session logs, metrics report, video capture, deck
Work packages	A 15 · B 12 · C 15 · D 13 (+ 8 shared, 7 gates)	A 17 · B 12 · C 17 · D 13	A 7 · B 8 · C 5 · D 8	A 5 · B 5 · C 6 · D 6

The lane owner decides inside the lane. Anything that touches a contract (§6.4) is announced to all four before it is committed.

1.3 How the plan is organised

plan §9.1

- **By dependency, not by date.** Every task lists what it needs and what proves it is done. A task starts when its inputs exist. A phase ends when its exit gate passes.
- **Contracts first.** The start is wide: three set-up tasks and four sets of contract pages have no dependency, so every engineer and their agents have work from the first minute. They meet in gate G0, where the interfaces between lanes are frozen and stubbed. After that the four lanes do not wait for each other's code until the integration gates.
- **Editor first, phone last.** Only one Mac can build to the phones, so everything that can be proved in the Unity Editor or in a unit test is proved there. The phone is for acceptance.
- **Two levels.** People plan in **work areas**: sixteen in Phase 1, a handful per lane, each a piece one person owns (§9.5.2); later phases have one to three areas per lane and use case (§9.6–9.8). Agents work on **tasks**: each task fits one coding agent, touches one lane's files, and has a binary acceptance test with numbers.
- **The plan's tables and diagrams come from one data set.** The Phase 1 graph has 70 tasks and 166 dependency edges, checked to be free of cycles; the diagrams below are generated from the same data as the tables.

1.4 Shared resources and the rules that protect them

plan §9.3

The build seat (one Mac).

1. Lane A's first jobs are the build chain (`P1-A02`, `P1-A13`) and the URP settings and the stereo rig (`P1-A03`, `P1-A04`). They take priority over every other lane-A task until G1 passes. Editor-only lane-A task `P1-A06` and the Editor part of `P1-A12` run alongside them as the graph in §9.5.5 allows.
2. Everyone runs the Unity Editor on their own machine. Native code is wrapped so the Editor and the Station build compile without it.
3. Device builds go out as a build train: one integration build at every gate, plus short on-request slots from a visible queue (who, branch, what is tested).
4. Prefer iteration that needs no rebuild: live tuning from the Console, replay of recorded RoomModels and pose logs in the Editor, kernel and graph logic as EditMode tests.
5. One bundle id and one signing account. A build signed by a different account cannot install over the existing one.

The two phones.

1. Phone A is the reference device for every published number. Phone B is lane C's working device (every one of lane C's Phase-1 tasks except the Editor-only FloorScan core, `P1-C15`, needs a phone), the hot-swap spare, and the second walker in Phase 3.
2. Hand-held before head-mounted: scan, link and input are debugged with the phone in the hand; the headset is for acceptance.
3. Record once, replay many times: the first good scan and a few walks are saved and replayed in the Editor by every lane.
4. A soak leaves a phone hot. Soaks are scheduled, not squeezed between someone else's measurements, and the thermal state is noted at the start of every measurement.

The Station's GPU. The RTX 5070 has 12 GB. The Forge's generator needs about 8 GB and owns the card while jobs are queued. The Spectator runs at modest settings beside it. The medical pipeline (`atlas case`, Phase 2) runs when the Forge is idle.

Scenes and prefabs. Unity scenes are single files, so each has exactly one owner.

Asset	Owner	Rule
`Atlas_Boot` (XR origin, stereo rig, SpaceLab, bootstraps)	A	Nobody else edits it. Systems register themselves from their own scenes
`Space_Home`	C	One additive scene per space, one owner each
`Space_Glued`, `Space_Portal`, `Space_Slicer`	A	
`Space_CurvedWalk`, `Space_AlignedCells`, `Space_SmallPlanet`	B	
`Sys_Scan`, `Sys_BumpGuard`, `Sys_Onboarding`	C	Loaded additively by the boot scene
`Sys_Link`, `Sys_Hud`, `PC_Spectator`	D	The Spectator scene is never in the iOS build list
Use-case scenes (Phase 2 onwards)	The lane that owns the use case's runtime	Reference spaces through `ISpaceModule`; never copy their objects
`Sandbox_<name>`	Each engineer	Never in the build list
URP assets, Project Settings, layers, stencil table, package manifest	A	Changed only by A, announced first
The shared material and palette	A	Per-tile and per-room material instances are created at run time from a pool, never saved as assets

The shared shader. Ownership of the four include files is in §8.3. A contract check runs at every gate: the shader is still reported SRP-Batcher compatible, and the draw-call count of the stress scene has not grown.

Source control. Text serialisation, visible meta files, UnityYAMLMerge as the merge tool, Git LFS for binaries. Small merges to one main branch. Tunables live in the parameter registry and in ScriptableObjects, one file per space, so tuning never touches a scene. Asset Store packs stay out of the repository.

2 Phase 1 · Foundation

plan §9.5

2.1 Who can work side by side

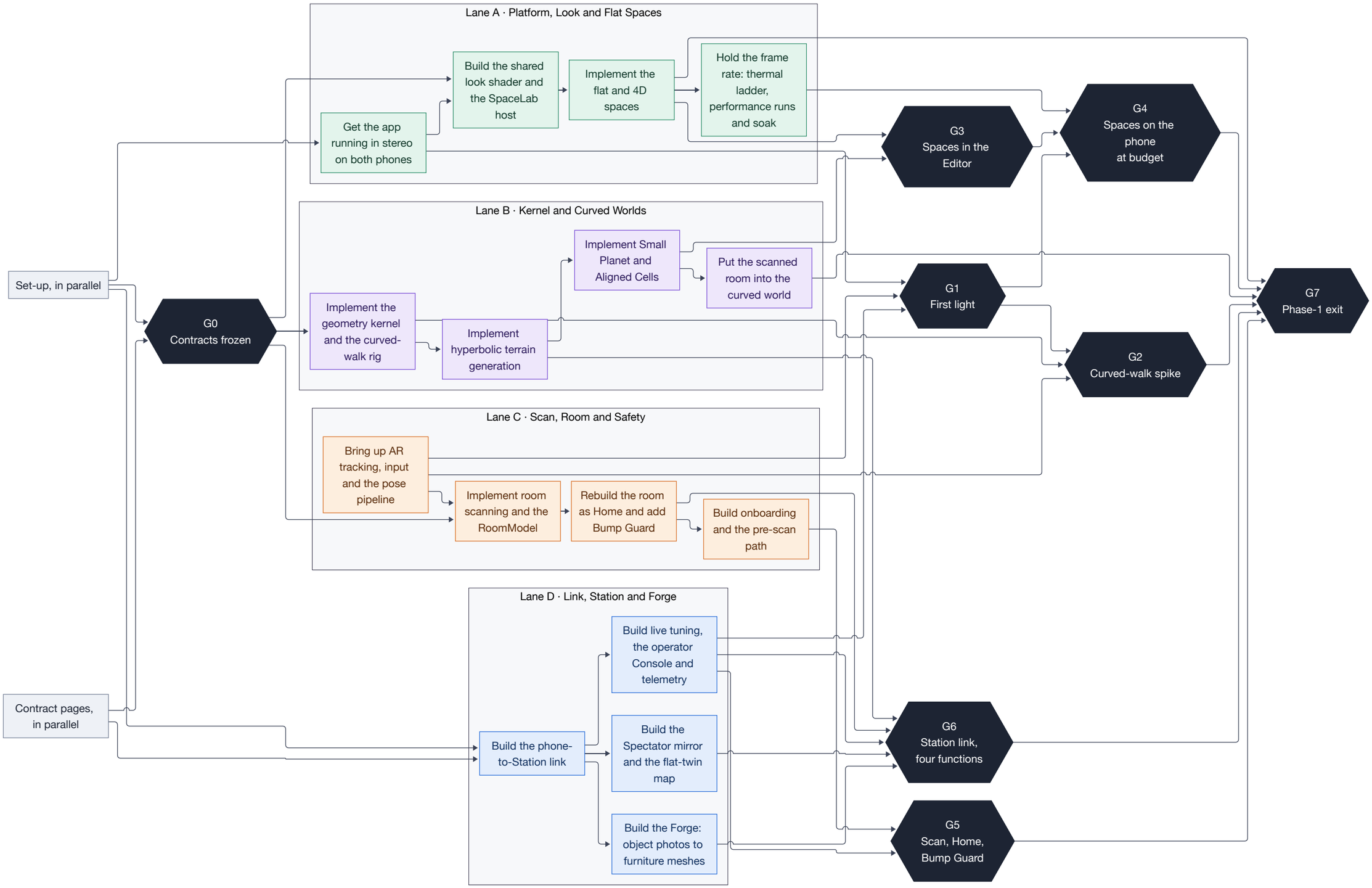
This phase opens the build. 17 areas cover the 63 tasks of this phase. Across a row: areas that can be worked on at the same point. Down a column: one person's order. An area sits in the first step in which part of it can begin; it is finished only when everything under "after" and "needs" has arrived.

Step	Lane A · Platform, Look and Flat Spaces	Lane B · Kernel and Curved Worlds	Lane C · Scan, Room and Safety	Lane D · Link, Station and Forge
1	**All** The start: set-up and contracts feeds G0			
2	**A1** Get the app running in stereo on both phones after All feeds G1	**B1** Implement the geometry kernel and the curved-walk rig after All needs G0 feeds G2	**C2** Implement room scanning and the RoomModel after All needs C1	**D1** Build the phone-to-Station link after All needs A1 feeds G1 **D4** Build the Forge: object photos to furniture meshes after All needs C3 D1 feeds G6
3	**A2** Build the shared look shader and the SpaceLab host after G0 A1	**B2** Implement hyperbolic terrain generation after B1 needs A2 feeds G3 G6	**C1** Bring up AR tracking, input and the pose pipeline after A1 needs A2 feeds G1 G2 G5	**D2** Build live tuning, the operator Console and telemetry after D1 needs C3 feeds G1 G5 G6
4	**A3** Implement the flat and 4D spaces after A2 needs C3 feeds G3 G7		**C3** Rebuild the room as Home and add Bump Guard after C1 needs C2 feeds G5 G6	**D3** Build the Spectator mirror and the flat-twin map after A2 D1 needs B2 feeds G6
5		**B3** Implement Small Planet and Aligned Cells after G0 B2 needs C3 feeds G3 G7	**C4** Build onboarding and the pre-scan path after C2 needs C3 D1 feeds G5	
6	**A4** Hold the frame rate: thermal ladder, performance runs and soak after C3 D2 needs G3 B4 feeds G4	**B4** Put the scanned room into the curved world after B2 C2 needs B3 feeds G7		

2.2 The work areas, as the plan draws them

plan §9.5.2

Sixteen work areas, a handful per lane. An area is the unit a person plans in ("hyperbolic world generation", "Bump Guard", "Forge: live furniture"); the tasks inside it (§9.5.4 onwards) are the units handed to agents. An arrow means the area at its head needs something from the area at its tail.



Drawn from the plan's diagram; each box shows the name of the area, and what it covers is in the tables that follow.

2.3 The areas, lane by lane

All hands

Area and what it delivers	Starts after	Also needs, before it can finish	Hands over to other lanes	Feeds gate	Inside, in the plan
All **The start: set-up and contracts** Build seat, signing and Unity (lane A leads, everyone takes part); Contracts frozen (this is gate G0); Station ready (lane D); Phones and network ready (lane C); Contract pages, lane A; Contract pages, lane B; Contract pages, lane C; Contract pages, lane D; Unity project and repository skeleton.	nothing in this phase	–	A1 Unity project and repository skeleton; Phones and network ready (lane C) B1 Unity project and repository skeleton C2 Unity project and repository skeleton; Contract pages, lane C D1 Contract pages, lane D; Station ready (lane D) D4 Station ready (lane D); Contract pages, lane D	G0	`9 tasks: P1–X01, X02, X03, X04, X05, X06, X07, X08, A01`

Lane A · Platform, Look and Flat Spaces

Area and what it delivers	Starts after	Also needs, before it can finish	Hands over to other lanes	Feeds gate	Inside, in the plan
A1 **Get the app running in stereo on both phones** the first build on both phones, URP settings and quality tiers, the two-camera stereo rig, the lens pre-distortion blit, and a one-command build	All	–	C1 First build on both phones; Two-camera stereo rig C3 Two-camera stereo rig D1 First build on both phones D2 Two-camera stereo rig	G1	`5 tasks: P1–A02, A03, A04, A05, A13`
A2 **Build the shared look shader and the SpaceLab host** the one shader and palette that every space uses, and the boot scene that loads one space at a time and runs the frame loop	G0 A1	–	B2 Shared look shader C1 SpaceLab host C3 Shared look shader D3 Shared look shader; SpaceLab host	–	`2 tasks: P1–A06, A08`
A3 **Implement the flat and 4D spaces** the Glued space (S1) with the scanned room in it, the stencil portal engine and Impossible Rooms (S5), and the Slicer (S6) with Mesh4D	A2	C3 Home rebuild S0	–	G3 G7	`5 tasks: P1–A09, A10, A11, A12, A15`

Area and what it delivers	Starts after	Also needs, before it can finish	Hands over to other lanes	Feeds gate	Inside, in the plan
A4 **Hold the frame rate: thermal ladder, performance runs and soak** the thermal quality ladder, a performance run for every space on the budget fixture, and the soak	C3 D2	G3 B4 The room becomes tile zero	–	G4	2 tasks: P1–A07, A14

Lane B · Kernel and Curved Worlds

Area and what it delivers	Starts after	Also needs, before it can finish	Hands over to other lanes	Feeds gate	Inside, in the plan
B1 **Implement the geometry kernel and the curved-walk rig** the double-precision kernel for hyperbolic, flat and spherical space, and the holonomy rig that turns head movement into movement through curved space	All	G0	–	G2	2 tasks: P1–B01, B02
B2 **Implement hyperbolic terrain generation** the warp shader, exact tile ids and the tile graph, procedural tile meshes, the tile renderer, and the world generator with WorldLint	B1	A2 Shared look shader	D3 Exact tile ids and the tile graph	G3 G6	5 tasks: P1–B03, B04, B05, B06, B07
B3 **Implement Small Planet and Aligned Cells** the spherical space (S4) and the flat-cell mode of the hyperbolic world (S3, Mode 2), with the scanned room in it	G0 B2	C3 Home rebuild S0	A4 Room content in Aligned Cells	G3 G7	3 tasks: P1–B08, B10, B12
B4 **Put the scanned room into the curved world** the curvature dial, the 3-4-5 walk, and the scanned room as tile zero of the curved world	B2 C2	B3 Small Planet S4	A4 The room becomes tile zero	G7	2 tasks: P1–B09, B11

Lane C · Scan, Room and Safety

Area and what it delivers	Starts after	Also needs, before it can finish	Hands over to other lanes	Feeds gate	Inside, in the plan
C1 **Bring up AR tracking, input and the pose pipeline** the AR session with Editor simulation and replay, gamepad and gaze input with twins, and the pose pipeline	A1	A2 SpaceLab host	D2 Input and twins	G1 G2 G5	3 tasks: P1–C01, C10, C11

Area and what it delivers	Starts after	Also needs, before it can finish	Hands over to other lanes	Feeds gate	Inside, in the plan
C2 **Implement room scanning and the RoomModel** the FloorScan core, the manual and automatic scan routes, object boxes, the RoomPlan wall bridge (Extended), and the RoomModel with the loop-fit	All	C1 AR session and Editor simulation	B4 RoomModel and the loop-fit	–	`6 tasks: P1–C02, C03, C04, C05, C06, C15`
C3 **Rebuild the room as Home and add Bump Guard** the Home rebuild (S0), object photo capture for the Forge, the warning cue and haptics, and Bump Guard	C1	C2 RoomModel and the loop-fit; Object boxes	A3 Home rebuild S0 A4 Home rebuild S0 B3 Home rebuild S0 B4 Home rebuild S0 D2 Warning cue and haptics D4 Home rebuild S0	G5 G6	`4 tasks: P1–C07, C08, C09, C14`
C4 **Build onboarding and the pre-scan path** onboarding with the phone in the hand, and a saved world map that relocalises in a room scanned earlier	C2	C3 Home rebuild S0 D1 Pairing and network bring-up	–	G5	`2 tasks: P1–C12, C13`

Lane D · Link, Station and Forge

Area and what it delivers	Starts after	Also needs, before it can finish	Hands over to other lanes	Feeds gate	Inside, in the plan
D1 **Build the phone-to-Station link** the Atlas.Link library, the Hub, and pairing and network bring-up	All	A1 First build on both phones	C4 Pairing and network bring-up	G1	`3 tasks: P1–D01, D02, D11`
D2 **Build live tuning, the operator Console and telemetry** the parameter registry and Console tuner, the operator Console, the on-device menu, and the perf probe, HUD and telemetry	D1	C3 Warning cue and haptics	A4 Perf probe, HUD and telemetry; Operator Console	G1 G5 G6	`4 tasks: P1–D03, D04, D06, D13`
D3 **Build the Spectator mirror and the flat-twin map** the Spectator build that re-renders the walker's view on the Station, and the flat-twin mini-map	A2 D1	B2 Exact tile ids and the tile graph	–	G6	`2 tasks: P1–D05, D07`
D4 **Build the Forge: object photos to furniture meshes** the Forge service on the Station, mesh clean-up, the asset client on the phone, and the built-in fallback set	All	C3 Home rebuild S0 D1 Atlas.Link library	–	G6	`4 tasks: P1–D08, D09, D10, D12`

2.4 Gates of this phase

plan §9.4

Gate	Needs	Passes when (summary)	Present
G0 Contracts frozen	All The start: set-up and contracts	Every contract of §6.4 is written and stubbed; each lane's sandbox runs against the stubs	all four lanes
G1 First light	A1 Get the app running in stereo on both phones C1 Bring up AR tracking, input and the pose pipeline D1 Build the phone-to-Station link D2 Build live tuning, the operator Console and telemetry G0	Empty stereo scene at 60 fps on the phone; telemetry visible on the Station; anyone can install a build	A, C, D
G2 Curved-walk spike	B1 Implement the geometry kernel and the curved-walk rig C1 Bring up AR tracking, input and the pose pipeline G1	Five-corner loop closes within 0.25 m and three walkers from the team are comfortable at the loop-fit curvature (with a spotter: Bump Guard does not exist yet). Decides the default mode and the comfort floor	B, C, three walkers from the team, a spotter
G3 Spaces in the Editor	A3 Implement the flat and 4D spaces B2 Implement hyperbolic terrain generation B3 Implement Small Planet and Aligned Cells G0	S1–S6 run in SpaceLab on the three sample rooms; all mathematics tests green	A, B
G4 Spaces on the phone at budget	A4 Hold the frame rate: thermal ladder, performance runs and soak G1 G3	Every space S0–S6 meets the Must column cold, with the room's content, on the `budget_room` fixture; the 15-minute soak passes	A, B, D
G5 Scan, Home and Bump Guard	C1 Bring up AR tracking, input and the pose pipeline C3 Rebuild the room as Home and add Bump Guard C4 Build onboarding and the pre-scan path D2 Build live tuning, the operator Console and telemetry G0	Both scan paths end in a registered Home; Bump Guard passes its scripted test in the headset	C, B, D
G6 Station link, four functions	B2 Implement hyperbolic terrain generation C3 Rebuild the room as Home and add Bump Guard D2 Build live tuning, the operator Console and telemetry D3 Build the Spectator mirror and the flat-twin map D4 Build the Forge: object photos to furniture meshes G0	Mirror, tuning and Console, telemetry, and the scan and asset round-trip all pass §7.6	D plus one engineer from each other lane
G7 Phase-1 exit	A3 Implement the flat and 4D spaces B3 Implement Small Planet and Aligned Cells B4 Put the scanned room into the curved world G2 G4 G5 G6	A first-time walker goes from self-scan to every space, watched and tuned from the Station	all four lanes

3 Phase 2 · Geometry Playground and Lung Mission

plan §9.6

3.1 Who can work side by side

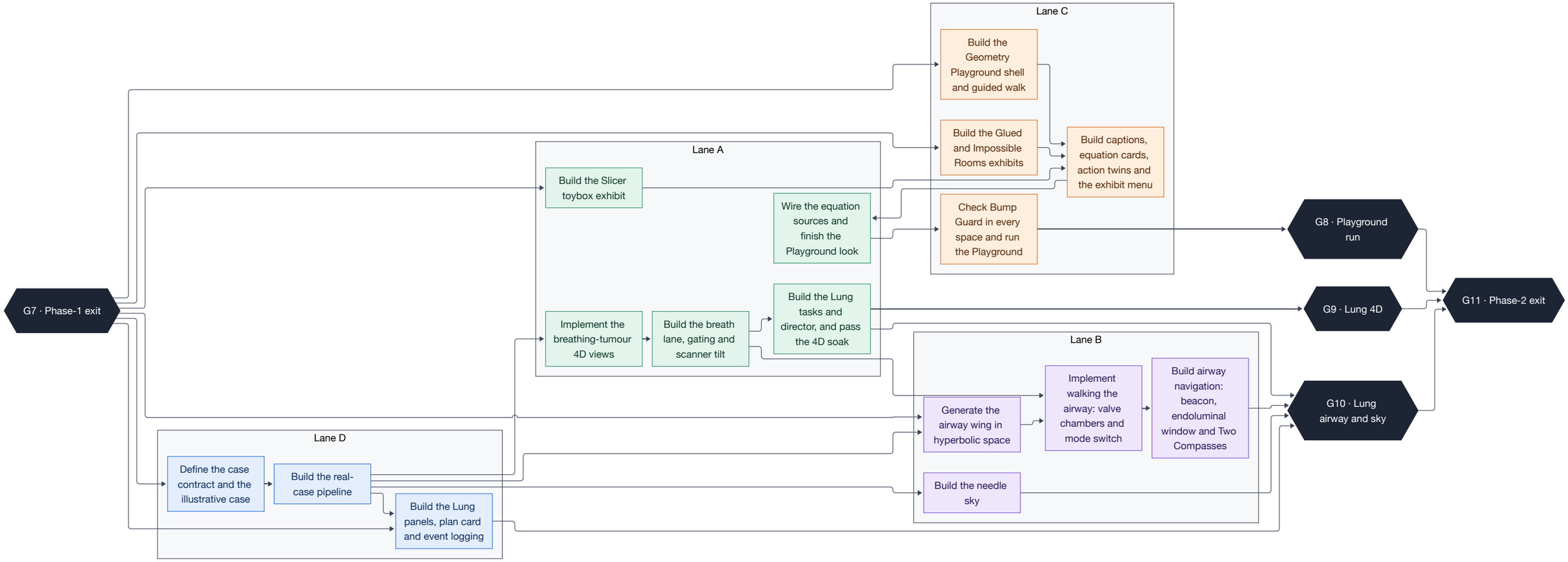
This phase opens when G7 Phase-1 exit has passed. 16 areas cover the 59 tasks of this phase. Across a row: areas that can be worked on at the same point. Down a column: one person's order. An area sits in the first step in which part of it can begin; it is finished only when everything under "after" and "needs" has arrived.

Step	Lane A · Platform, Look and Flat Spaces	Lane B · Kernel and Curved Worlds	Lane C · Scan, Room and Safety	Lane D · Link, Station and Forge
1	A1 Build the Slicer toybox exhibit	B1 Generate the airway wing in hyperbolic space needs D2	C1 Build the Geometry Playground shell and guided walk	D1 Define the case contract and the illustrative case
			C3 Build the Glued and Impossible Rooms exhibits	D3 Build the Lung panels, plan card and event logging needs D2 feeds G10
2	A2 Wire the equation sources and finish the Playground look after A1 C3 needs C2		C2 Build captions, equation cards, action twins and the exhibit menu after C1 needs A1 C3	D2 Build the real-case pipeline after D1
	A3 Implement the breathing-tumour 4D views after D1 needs D2			
3	A4 Build the breath lane, gating and scanner tilt after A3	B3 Build airway navigation: beacon, endoluminal window and Two Compasses after D2 needs B2 feeds G10	C4 Check Bump Guard in every space and run the Playground after A2 feeds G8	
		B4 Build the needle sky after D2 feeds G10		
4	A5 Build the Lung tasks and director, and pass the 4D soak after A4 feeds G9 G10	B2 Implement walking the airway: valve chambers and mode switch after B1 needs A4 feeds G10		

3.2 The work areas, as the plan draws them

plan §9.6

The areas are for people; the packages inside them are for agents. An arrow from a gate into an area means work in that area can start at that gate; an arrow between areas means the second cannot be finished without the first.



Colours: green lane A, purple lane B, orange lane C, blue lane D; dark hexagons are gates. Drawn from the plan's diagram; each box shows the name of the area, and what it covers is in the tables that follow.

3.3 The areas, lane by lane

Lane A · Platform, Look and Flat Spaces

Area and what it delivers	Starts after	Also needs, before it can finish	Hands over to other lanes	Feeds gate	Inside, in the plan
A1 **Build the Slicer toybox exhibit** the Slicer toybox with section names and shadow	nothing in this phase	–	C2 Slicer toybox	–	`2 tasks: P2-A01, A02`
A2 **Wire the equation sources and finish the Playground look** the live numbers behind the equation cards of the exhibits, and the look polish of the Playground	A1 C3	C2 Caption chips and equation cards	C4 Look polish; Exhibit equation sources	–	`2 tasks: P2-A03, A04`
A3 **Implement the breathing-tumour 4D views** the 4D tube loaded into Mesh4D, live slices with phase colouring, shells and the look-down shadow, the breathing chest, and the ring fallback	D1	D2 atlas case chest	–	–	`6 tasks: P2-A20, A21, A22, A23, A24, A30`
A4 **Build the breath lane, gating and scanner tilt** the breath lane with the unroll animation and "now" plane, the gating flags, and the scanner tilt	A3	–	B2 Unroll animation and "now" plane	–	`4 tasks: P2-A25, A26, A27, A28`
A5 **Build the Lung tasks and director, and pass the 4D soak** tasks M2, M3 and M4 with scoring and events, the Lung director with labels and credits, and the Lung 4D soak report	A4	–	–	G9 G10	`3 tasks: P2-A29, A31, A32`

Lane B · Kernel and Curved Worlds

Area and what it delivers	Starts after	Also needs, before it can finish	Hands over to other lanes	Feeds gate	Inside, in the plan
B1 **Generate the airway wing in hyperbolic space** the airway tree mapped onto the tile graph, and the corridor and chamber kit	nothing in this phase	D2 atlas case airway	–	–	`2 tasks: P2-B01, B02`
B2 **Implement walking the airway: valve chambers and mode switch** the valve solver for Mode 1, the valve walk for Mode 2, the mode switch, and the lesion chamber hand-over	B1	A4 Unroll animation and "now" plane	–	G10	`4 tasks: P2-B03, B04, B05, B09`

Area and what it delivers	Starts after	Also needs, before it can finish	Hands over to other lanes	Feeds gate	Inside, in the plan
B3 **Build airway navigation: beacon, endoluminal window and Two Compasses** the beacon and route cost, the endoluminal window, Two Compasses, and the airway run test	D2	B2 Mode switch	–	G10	`4 tasks: P2-B06, B07, B08, B12`
B4 **Build the needle sky** the needle sky with its phase and robust layers, and the great-circle check	D2	–	–	G10	`2 tasks: P2-B10, B11`

Lane C · Scan, Room and Safety

Area and what it delivers	Starts after	Also needs, before it can finish	Hands over to other lanes	Feeds gate	Inside, in the plan
C1 **Build the Geometry Playground shell and guided walk** the director and beats, scan-to-Home, the centre mark and dial hand-off, the guided 3-4-5 walk, the mode compare, tutorial prompts with the comfort question, and the pointing probe	nothing in this phase	–	–	–	`7 tasks: P2-C01, C02, C03, C04, C05, C09, C17`
C2 **Build captions, equation cards, action twins and the exhibit menu** caption chips and equation cards, the action twins, and the exhibit menu with return to Home	C1	A1 Slicer toybox C3 Glued exhibit: wrap-walk; Impossible Rooms exhibit	A2 Caption chips and equation cards	–	`3 tasks: P2-C06, C07, C08`
C3 **Build the Glued and Impossible Rooms exhibits** the Glued exhibit with the chiral kit and the gluing switch, and the Impossible Rooms exhibit with the loop corridor	nothing in this phase	–	A2 Glued exhibit: wrap-walk; Impossible Rooms exhibit; Chiral kit; Gluing switch and T^3; Loop corridor and room numbers	–	`5 tasks: P2-C12, C13, C14, C15, C16`
C4 **Check Bump Guard in every space and run the Playground** the Bump Guard check repeated in every space, and the complete Playground run	A2	–	–	G8	`2 tasks: P2-C10, C11`

Lane D · Link, Station and Forge

Area and what it delivers	Starts after	Also needs, before it can finish	Hands over to other lanes	Feeds gate	Inside, in the plan
D1 **Define the case contract and the illustrative case** the case contract with its readers, and the illustrative case package that lanes A and B build against	nothing in this phase	–	A3 Case contract; atlas case twin	–	`2 tasks: P2-D01, D02`
D2 **Build the real-case pipeline** the `atlas case` steps from fetch and masks to the 4D tube, shells, chest, airway and sky, packed and delivered to the phone	D1	–	A3 atlas case chest A5 atlas case pack and delivery B1 atlas case airway B3 atlas case airway B4 atlas case sky	–	`7 tasks: P2-D03, D04, D05, D06, D07, D08, D09`
D3 **Build the Lung panels, plan card and event logging** the Lung Console panel, the Lung twins on Spectator and Console, the plan card, and event logging with the pilot export	nothing in this phase	D2 atlas case airway	–	G10	`4 tasks: P2-D10, D11, D12, D13`

3.4 Gates of this phase

plan §9.4

Gate	Needs	Passes when (summary)	Present
G8 **Playground run**	C4 Check Bump Guard in every space and run the Playground G7	A first-time walker completes the guided run in two different rooms	A, C, two first-time users
G9 **Lung 4D**	A5 Build the Lung tasks and director, and pass the 4D soak G7	The four 4D pieces run on both case packages at budget	A, B, D
G10 **Lung airway and sky**	A5 Build the Lung tasks and director, and pass the 4D soak B2 Implement walking the airway: valve chambers and mode switch B3 Build airway navigation: beacon, endoluminal window and Two Compasses B4 Build the needle sky D3 Build the Lung panels, plan card and event logging G7	Four forks walked without rescue; sky and plan card correct	B, D
G11 **Phase-2 exit**	G8 G9 G10	Playground and Lung Mission in one session by an outsider, hot, at budget	all four lanes

3.5 What the plan says about this phase

plan §9.6

Two tracks run in parallel from G7. The specifications and the work-package tables are in §10.1 (Playground) and §10.2 (Lung Mission); the diagrams here show the order.

Track	Lanes	Packages	Exit gate
Geometry Playground	C (shell; Glued and Impossible Rooms exhibits) and A (Slicer toybox, equation sources, look)	`P2-C01`–`P2-C17`, `P2-A01`–`P2-A04`	G8
Lung Mission	D (Station pipeline, case packages, plan card), A (medical 4D runtime), B (airway wing, needle sky)	`P2-D01`–`P2-D13`, `P2-A20`–`P2-A32`, `P2-B01`–`P2-B12`	G9 (4D), G10 (airway and sky)

Lane A serves both tracks, which departs from a strict "two lanes each" (open item 17 in §14): the Slicer toybox and the medical 4D runtime both build on lane A's `Mesh4D`. Lane A does `P2-A01` and `P2-A02` first, because `P2-A01` unblocks lane C's exhibit menu (`P2-C08`), then starts `P2-A20` onwards; `P2-A03` and `P2-A04` follow when their lane-C inputs land.

What is sequential:

- In the Playground everything hangs off the director (`P2-C01`); the exhibits wrap Phase 1 spaces and are independent of one another; the run (`P2-C11`) is last.
- In the Lung Mission the case contract (`P2-D01`) comes first. Lane D then produces the **illustrative** package (`P2-D02`) before the real one, so lanes A and B have data to build against without waiting for the full pipeline. The 4D runtime (lane A) and the airway wing (lane B) are independent of each other until the lesion chamber hand-over (`P2-B09`). The Lung director (`P2-A32`) stubs the beats that belong to lanes B and D, so G9 does not wait for the airway.
- G11 needs G8, G9 and G10, and then one combined session by someone from outside the team.

4 Phase 3 · Shared House and FieldFit

plan §9.7

4.1 Who can work side by side

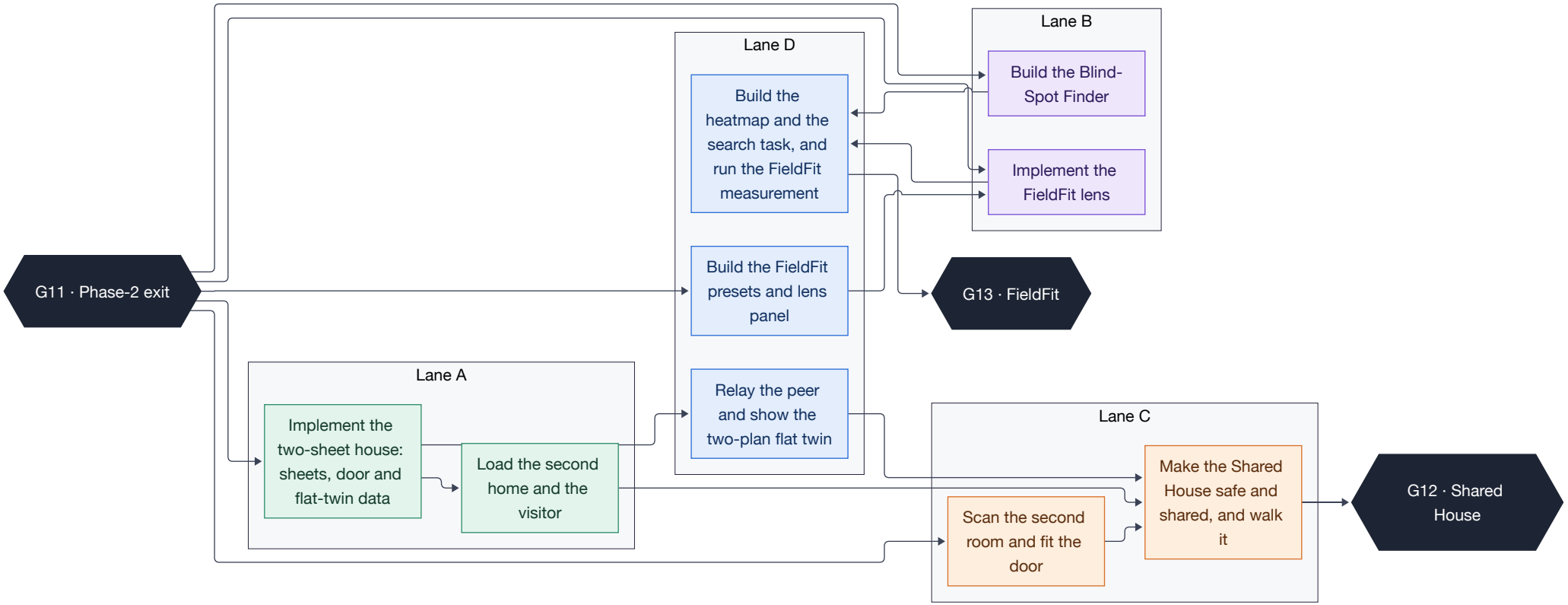
This phase opens when G11 Phase-2 exit has passed. 9 areas cover the 28 tasks of this phase. Across a row: areas that can be worked on at the same point. Down a column: one person's order. An area sits in the first step in which part of it can begin; it is finished only when everything under "after" and "needs" has arrived.

Step	Lane A · Platform, Look and Flat Spaces	Lane B · Kernel and Curved Worlds	Lane C · Scan, Room and Safety	Lane D · Link, Station and Forge
1	A1 **Implement the two-sheet house: sheets, door and flat-twin data**	B1 **Implement the FieldFit lens** needs D2 B2 **Build the Blind-Spot Finder**	C1 **Scan the second room and fit the door**	D1 **Relay the peer and show the two-plan flat twin** needs A1 D2 **Build the FieldFit presets and lens panel** needs B1
2	A2 **Load the second home and the visitor** after A1 needs C1			D3 **Build the heatmap and the search task, and run the FieldFit measurement** after B1 needs B2 feeds G13
3			C2 **Make the Shared House safe and shared, and walk it** after A2 needs D1 feeds G12	

4.2 The work areas, as the plan draws them

plan §9.7

An arrow from a gate into an area means work in that area can start at that gate; an arrow between areas means the second cannot be finished without the first.



Colours: green lane A, purple lane B, orange lane C, blue lane D; dark hexagons are gates. Drawn from the plan's diagram; each box shows the name of the area, and what it covers is in the tables that follow.

4.3 The areas, lane by lane

Lane A · Platform, Look and Flat Spaces

Area and what it delivers	Starts after	Also needs, before it can finish	Hands over to other lanes	Feeds gate	Inside, in the plan
A1 **Implement the two-sheet house: sheets, door and flat-twin data** the sheet state machine, the two-sheet door rendering, and the winding counter with the flat-twin data	nothing in this phase	–	D1 Winding counter and flat-twin data	–	`3 tasks: P3-A01, A02, A05`
A2 **Load the second home and the visitor** the second home loader, the visitor avatar with its recorded track, the phone performance pass, and the guest rug if the lead approves it	A1	C1 Door placement and door fit	C2 Second home loader; Visitor avatar, recorded track, labels; Phone performance pass	–	`4 tasks: P3-A03, A04, A06, A07`

Lane B · Kernel and Curved Worlds

Area and what it delivers	Starts after	Also needs, before it can finish	Hands over to other lanes	Feeds gate	Inside, in the plan
B1 **Implement the FieldFit lens** the lens mathematics, the lookup builder, the wide source render with the composed final blit, the field-loss mask and slider, the lens guard, and the phone performance pass	nothing in this phase	D2 Field presets	D2 Lookup builder D3 Lens maths library; Field-loss mask and slider; Phone performance pass	–	`6 tasks: P3-B01, B02, B03, B04, B05, B08`
B2 **Build the Blind-Spot Finder** the Blind-Spot Finder, and Scotoma River (Extended)	nothing in this phase	–	D3 Blind-Spot Finder	–	`2 tasks: P3-B06, B07`

Lane C · Scan, Room and Safety

Area and what it delivers	Starts after	Also needs, before it can finish	Hands over to other lanes	Feeds gate	Inside, in the plan
C1 **Scan the second room and fit the door** the second scan path, and door placement with the door fit	nothing in this phase	–	A2 Door placement and door fit	–	`2 tasks: P3-C01, C02`
C2 **Make the Shared House safe and shared, and walk it** Bump Guard on the second sheet, the peer link (Extended), and the G12 walk-through	A2	D1 Peer relay (Extended); Two-plan flat twin	–	G12	`3 tasks: P3-C03, C04, C05`

Lane D · Link, Station and Forge

Area and what it delivers	Starts after	Also needs, before it can finish	Hands over to other lanes	Feeds gate	Inside, in the plan
D1 **Relay the peer and show the two-plan flat twin** the peer relay (Extended) and the two floor plans with the winding dial on the Station	nothing in this phase	A1 Winding counter and flat-twin data	C2 Peer relay (Extended); Two-plan flat twin	–	`2 tasks: P3-D07, D08`
D2 **Build the FieldFit presets and lens panel** the field presets, the Console lens panel, and the Teichmüller lookup (Stretch)	nothing in this phase	B1 Lookup builder	B1 Field presets	–	`3 tasks: P3-D01, D02, D05`
D3 **Build the heatmap and the search task, and run the FieldFit measurement** the Spectator heatmap and bound, the search task with logging, and the G13 measurement run	B1	B2 Blind-Spot Finder	–	G13	`3 tasks: P3-D03, D04, D06`

4.4 Gates of this phase

plan §9.4

Gate	Needs	Passes when (summary)	Present
G12 **Shared House**	C2 Make the Shared House safe and shared, and walk it G11	Post Circling walked by three people with no contact	A, C, both phones
G13 **FieldFit**	D3 Build the heatmap and the search task, and run the FieldFit measurement G11	Blind spot found; search is faster with the lens than without	B, D

4.5 What the plan says about this phase

plan §9.7

> Two independent use cases, two lanes each; lane D adds two small packages to the Shared House (the peer relay and the two-plan flat twin). Specifications and work-package tables: §10.3 and §10.4. Packages marked Extended, Stretch or Blocked are drawn but feed no gate.

5 Phase 4 · Finish

plan §9.8

5.1 Who can work side by side

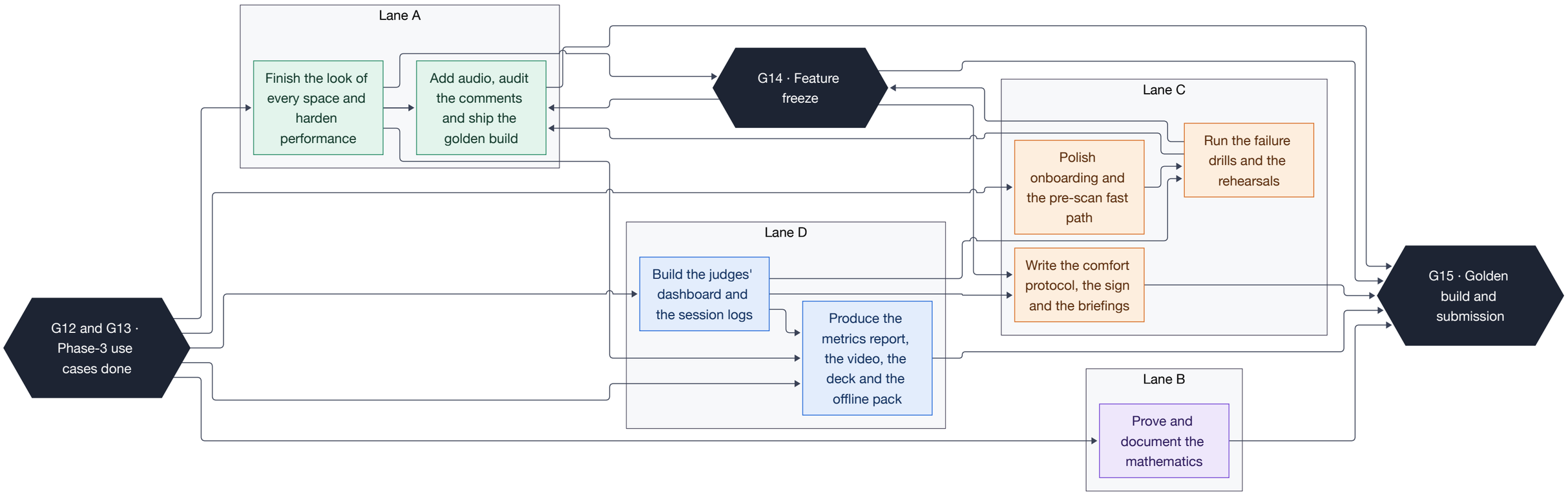
This phase opens when G12 Shared House and G13 FieldFit have passed. 8 areas cover the 22 tasks of this phase. Across a row: areas that can be worked on at the same point. Down a column: one person's order. An area sits in the first step in which part of it can begin; it is finished only when everything under "after" and "needs" has arrived.

Step	Lane A · Platform, Look and Flat Spaces	Lane B · Kernel and Curved Worlds	Lane C · Scan, Room and Safety	Lane D · Link, Station and Forge
1	**A1** Finish the look of every space and harden performance feeds G14	**B1** Prove and document the mathematics feeds G15	**C1** Polish onboarding and the pre-scan fast path feeds G15	**D1** Build the judges' dashboard and the session logs **D2** Produce the metrics report, the video, the deck and the offline pack needs A1 D1 feeds G15
2			**C2** Run the failure drills and the rehearsals after C1 needs D1 feeds G14 G15 **C3** Write the comfort protocol, the sign and the briefings needs G14 feeds G15	
3	**A2** Add audio, audit the comments and ship the golden build after A1 C2 needs G14 feeds G15			

5.2 The work areas, as the plan draws them

plan §9.8

An arrow from a gate into an area means work in that area can start at that gate; an arrow between areas means the second cannot be finished without the first.



Colours: green lane A, purple lane B, orange lane C, blue lane D; dark hexagons are gates. Drawn from the plan’s diagram; each box shows the name of the area, and what it covers is in the tables that follow.

5.3 The areas, lane by lane

Lane A · Platform, Look and Flat Spaces

Area and what it delivers	Starts after	Also needs, before it can finish	Hands over to other lanes	Feeds gate	Inside, in the plan
A1 **Finish the look of every space and harden performance** the look pass for every space and the soak on both phones, which is what the feature freeze waits for	nothing in this phase	–	D2 Performance hardening	G14	`2 tasks: P4-A01, A03`
A2 **Add audio, audit the comments and ship the golden build** the audio loops and cues, the comment audit, and the golden build with its launch check	A1 C2	G14	–	G15	`3 tasks: P4-A02, A04, A05`

Lane B · Kernel and Curved Worlds

Area and what it delivers	Starts after	Also needs, before it can finish	Hands over to other lanes	Feeds gate	Inside, in the plan
B1 **Prove and document the mathematics** the maths notes, the correctness report, the equation-card audit, the figure data, and the prior-art note	nothing in this phase	–	–	G15	`5 tasks: P4-B01, B02, B03, B04, B05`

Lane C · Scan, Room and Safety

Area and what it delivers	Starts after	Also needs, before it can finish	Hands over to other lanes	Feeds gate	Inside, in the plan
C1 **Polish onboarding and the pre-scan fast path** onboarding polished with outside walkers, and the pre-scan fast-path drill	nothing in this phase	–	–	G15	`2 tasks: P4-C01, C03`
C2 **Run the failure drills and the rehearsals** the failure drills and the rehearsals with outside walkers	C1	D1 Judges' dashboard	A2 Failure drills	G14 G15	`2 tasks: P4-C04, C05`
C3 **Write the comfort protocol, the sign and the briefings** the comfort protocol, the booth sign and the spotter briefing	nothing in this phase	G14	–	G15	`2 tasks: P4-C02, C06`

Lane D · Link, Station and Forge

Area and what it delivers	Starts after	Also needs, before it can finish	Hands over to other lanes	Feeds gate	Inside, in the plan
D1 **Build the judges' dashboard and the session logs** the judges' dashboard, and the session log with the pilot data	nothing in this phase	–	C2 Judges' dashboard C3 Session log and pilot data	–	`2 tasks: P4-D01, D02`
D2 **Produce the metrics report, the video, the deck and the offline pack** the metrics report, the Spectator capture, the deck with its claims check, and the offline pack	nothing in this phase	A1 Performance hardening D1 Session log and pilot data; Judges' dashboard	–	G15	`4 tasks: P4-D03, D04, D05, D06`

5.4 Gates of this phase

plan §9.4

Gate	Needs	Passes when (summary)	Present
G14 **Feature freeze**	A1 Finish the look of every space and harden performance C2 Run the failure drills and the rehearsals G12 G13	Full path twice in a row on one phone without cooling; only fixes after this	all four lanes
G15 **Golden build and submission**	A2 Add audio, audit the comments and ship the golden build B1 Prove and document the mathematics C1 Polish onboarding and the pre-scan fast path C2 Run the failure drills and the rehearsals C3 Write the comfort protocol, the sign and the briefings D2 Produce the metrics report, the video, the deck and the offline pack G14	Golden build runs with the Station unplugged; documents, video and metrics delivered	all four lanes

5.5 What the plan says about this phase

plan §9.8

> Polish, evaluation, demo operations and documentation. Work-package table: §13.6. Feature freeze (G14) follows the look pass, the performance hardening and the failure drills; after it, only fixes, audio, documents and rehearsals. In the diagrams of §9.7 and §9.8 the start gate shows when a lane is free; the tables give the technical dependency (§9.4).

6 Working with AI coding agents

plan §9.9

The tasks are sized and worded to be handed over. The rules that make that safe:

1. **One task, one branch, one merge request.** The branch name is the task id.
2. **An agent stays in its lane's folders.** It may read contracts and other lanes' public interfaces; it may not edit them. A contract change is made by a human and announced.
3. **The acceptance test comes first.** For anything provable in the Editor the agent writes the test from the acceptance cell, sees it fail, then implements.
4. **Parallel agents take tasks marked ll** in §9.5.1: they touch different files by construction.
5. **A human runs every phone test** and records the numbers. An agent prepares the build, the test scene and the checklist.
6. **No agent merges on a red test or a broken shader contract.**
7. **Numbers in a task are part of the task.** If a budget cannot be met, the agent reports the measured value and stops; it does not relax the number.

A task brief, ready to paste:

```
Task        P1-B04  Exact tile ids and the tile graph           Lane B
Read first  docs/contracts/kernel.md; this plan §3.4, §6.5, Appendix A.5
Depends on  P1-B01 (merged)
You own     Assets/Atlas/Kernel/Tiling/**  and its tests. Touch nothing else.
Deliver     TileId.cs, TileGraph.cs, Tests/TileIdTests.cs
Accept      - per-depth tile counts equal OEIS A377322 to depth 10
            - a walk round a vertex closes after exactly 5 steps
            - φ(X²+Y²) - T² = -1 for every generated tile (evaluated in Int128)
            - tile centres within ρ = 2 / 3 / 3.5 / 4 of a tile centre: 13 / 45 / 81 / 129; within 3.8425: 105
            - a tile crossing costs ≤ 0.5 ms in the Editor
Out of scope  rendering, materials, scenes, shader code
Report      test log; measured crossing cost; anything in the contract you could not satisfy
```